

# Database Administration

## Lab 02: Reverse Design

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August 10, 2026

## 1 Introduction

The main objective of this lab is to explore a database tool to work on the DBA database design, specifically with the technique of Reverse Design to obtain an ERD. **DBeaver** is a multiplatform DBA tool designed for diverse DBMSs. It is a Java-based application with a GUI that can connect to a database (via a JDBC driver), extract the required data, and generate an ERD from scratch. We will cover a basic guide using a couple of DBMSs, complete some sections of the online course, and then explore other plausible alternatives as autonomous work.

## 2 DBeaver Tutorial

### 2.1 Database Server Requirement

These exercises assume you have a functional instance of **PostgreSQL** running.

- **Lab Computers:** Follow the Vagrant setup instructions provided (Lab00).
- **Personal Machines:** Ensure PostgreSQL is installed and active. If you already have a running installation on your system, you may skip the server configuration steps.

### 2.2 Installing DBeaver

To interact with the database, we will use the DBeaver GUI. You can download the appropriate installer for your operating system from the official download page: <https://dbeaver.io/download/>.

- **Windows & macOS:** Download and follow the standard installation wizard.
- **Linux:** Download the Linux package and uncompress the file, and you are ready to go! So let's follow these steps:

1. Open a terminal console and type:

```
wget https://dbeaver.io/files/dbeaver-ce-latest-linux.gtk.x86_64.tar.gz
```

2. Decompress the tar file by executing:

```
tar -zxvf dbeaver-ce-latest-linux.gtk.x86_64.tar.gz
```

3. Now, you can execute DBeaver from your terminal by navigating inside the dbeaver folder and running the executable. Type the following commands:

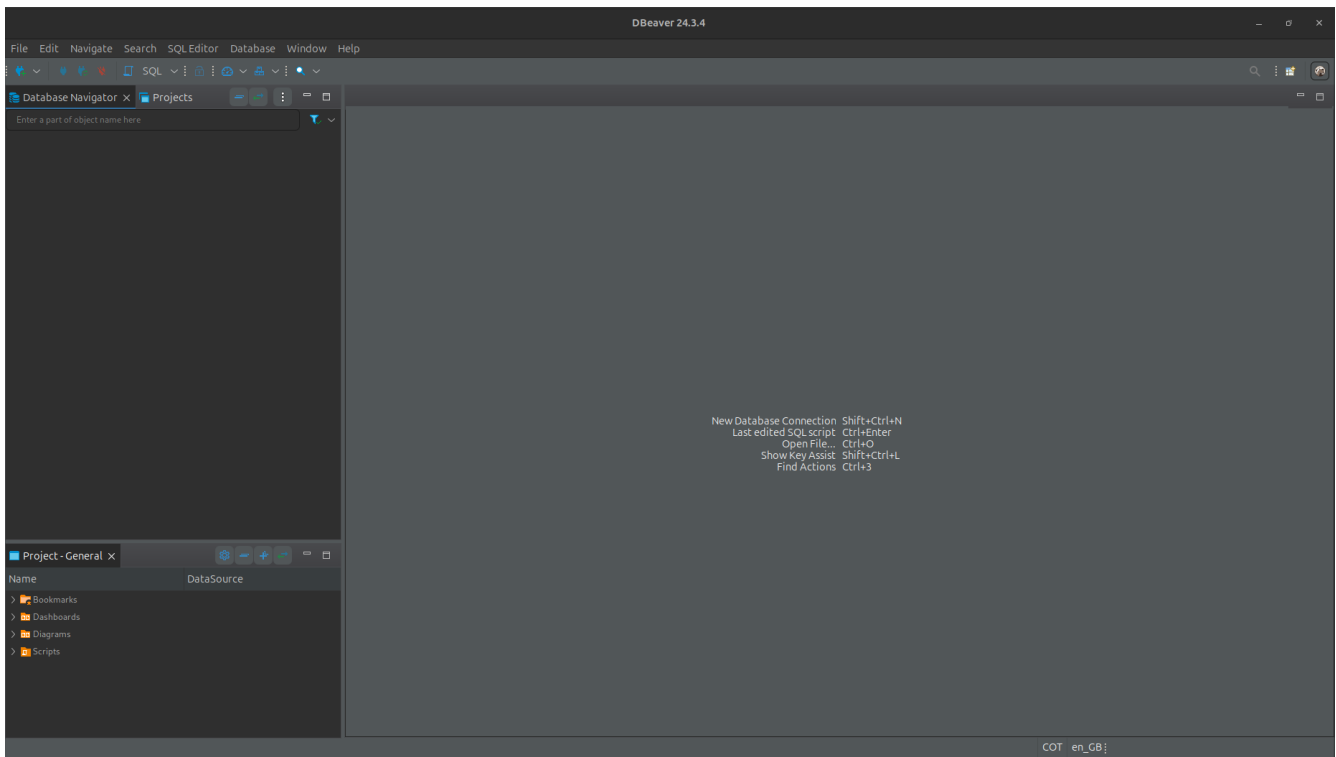


Figure 1: DBeaver GUI.

```
cd dbeaver
./dbeaver
```

If everything goes well, you will see something like Figure 1.

### 2.3 Reverse ERD using SQLite

Overall, this will be very simple. We will connect to a sample database called Chinook, implemented in [SQLite3](#). SQLite is a highly portable DBMS, and you will not need any additional installation (I believe?). You can download the database from [here](#).

Follow this [video](#) to explore the steps for generating the ERD for this database by making the connection from DBeaver. Let's go!

### 2.4 Reverse ERD using PostgreSQL

Ok, now we will give our PostgreSQL installation a try with an interesting sample database named [Pagila](#). You can download the database schema from [here](#) and the actual data from [here](#).

Again, follow this new [video](#) to explore the steps for generating the ERD, this time by connecting to a PostgreSQL instance. Here we go!

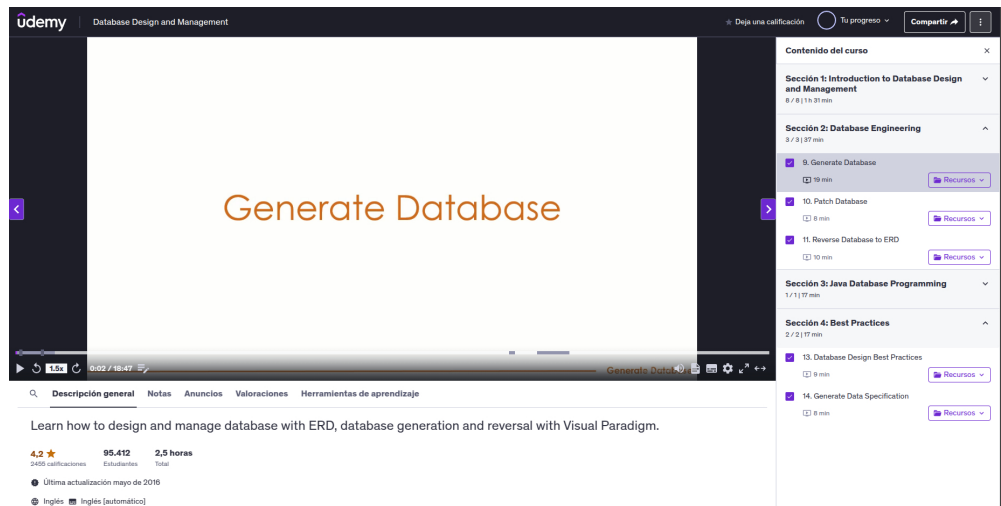


Figure 2: Progress in the *Database Design and Management* Udemy course (checkboxes on the right).

## 3 Autonomous Work

### 3.1 Complete Others Sections in the Course

You are asked to create an account on Udemy and visit the webpage of the *Database Design and Management* course. Follow sections 2 and 4. Once you complete those sections, please take a screenshot of your progress (see Figure 2 for an example) and attach it to your report.

### 3.2 Explore Others Options

As before, we will now explore the same topic using a similar approach but this time by trying out online tools for creating E-R diagrams. Read the list of a few interesting tools [here](#).

So, you will have a list of E-R Diagram tools:

- Azimutt
- draw.io
- Lucidchart
- QuickDBD
- DBDiagram.io
- DrawSQL
- Dia

and a list of DBMSs:

- Oracle
- SQLServer
- MySQL

- DB2

Now you will choose a combination of a E-R Diagram tool and a DBMS (i.e. Lucidchart + MySQL) and then write a well-structured tutorial on how to use that tool to generate or create ERDs for the **Chinook** database for that particular DBMS.

We expect you to submit your report by **August 24, 2026**, via the link that will be provided on Brightspace™.

Happy Hacking! 😎